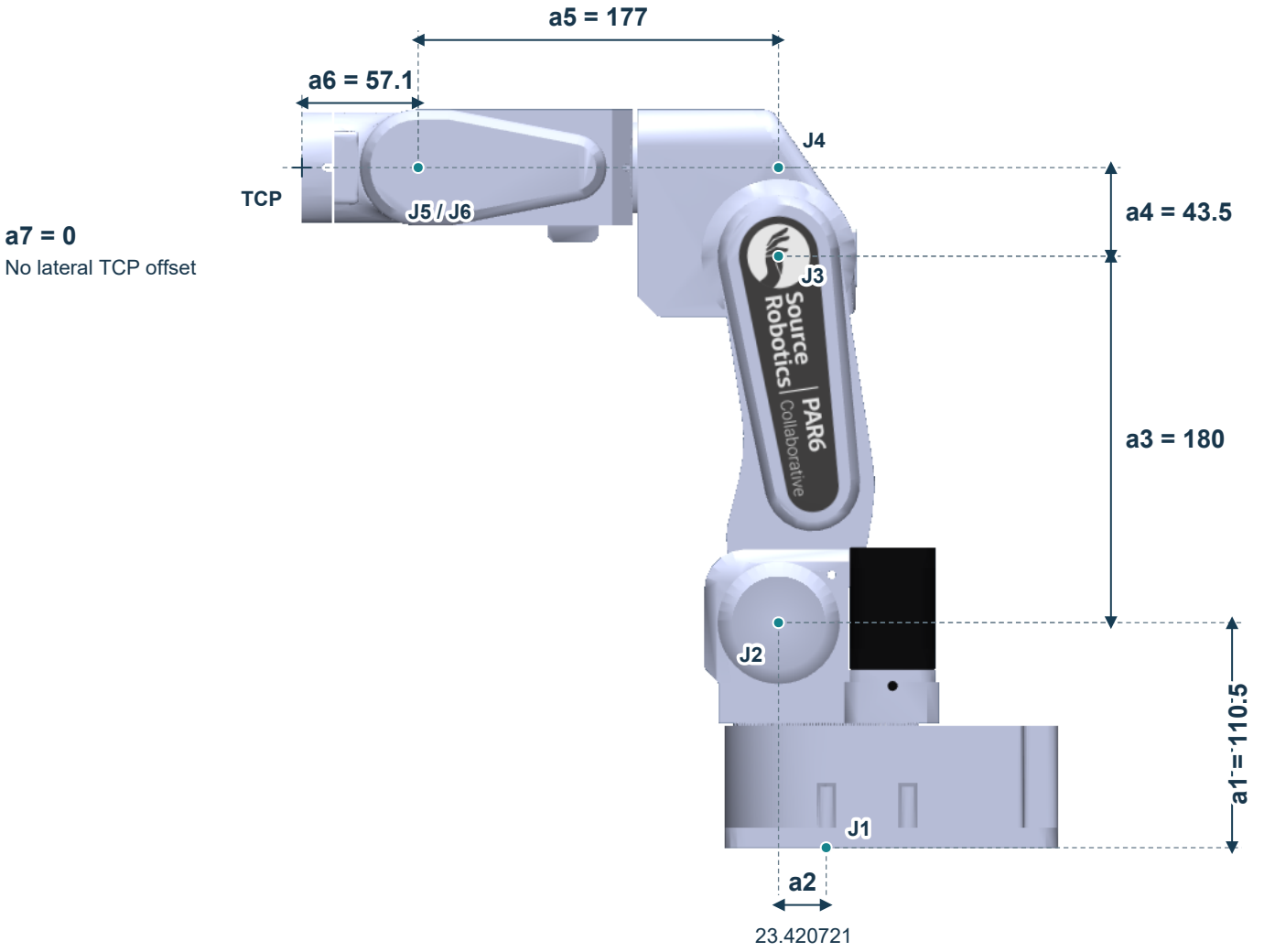
# PAR6 / Flange only

## Kinematic review | Standard DH to the configured TCP



Pose: q1 ... q6 = 0 rad; no jaws. Orthographic side view; dimensions in mm.




a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

| Joint | Link | theta [rad] | alpha [deg] | a [mm] | d [mm] |
|---|---|---|---|---|---|
| 1 | Base | q1 | -90 | a2 = 23.420721 | a1 = 110.5 |
| 2 | Shoulder | q2 - pi/2 | 180 | a3 = 180 | 0 |
| 3 | Elbow | q3 + pi | 90 | -a4 = -43.5 | 0 |
| 4 | Wrist 1 | q4 | -90 | 0 | -a5 = -177 |
| 5 | Wrist 2 | q5 | 90 | 0 | 0 |
| 6 | Wrist 3 / TCP | q6 + pi | 180 | -a7 = 0 | -a6 = -57.1 |

TCP is the configured flange reference. No gripper is installed.
a6 measures J6 center to flange; a7 is zero.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.004048 mm / 0.001350 deg
Max ideal DH vs URDF: 0.000000 mm / 0.000000 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

Source: par6_flange.urdf + scene_flange.xml | Full transforms, hashes and checks: flange.json

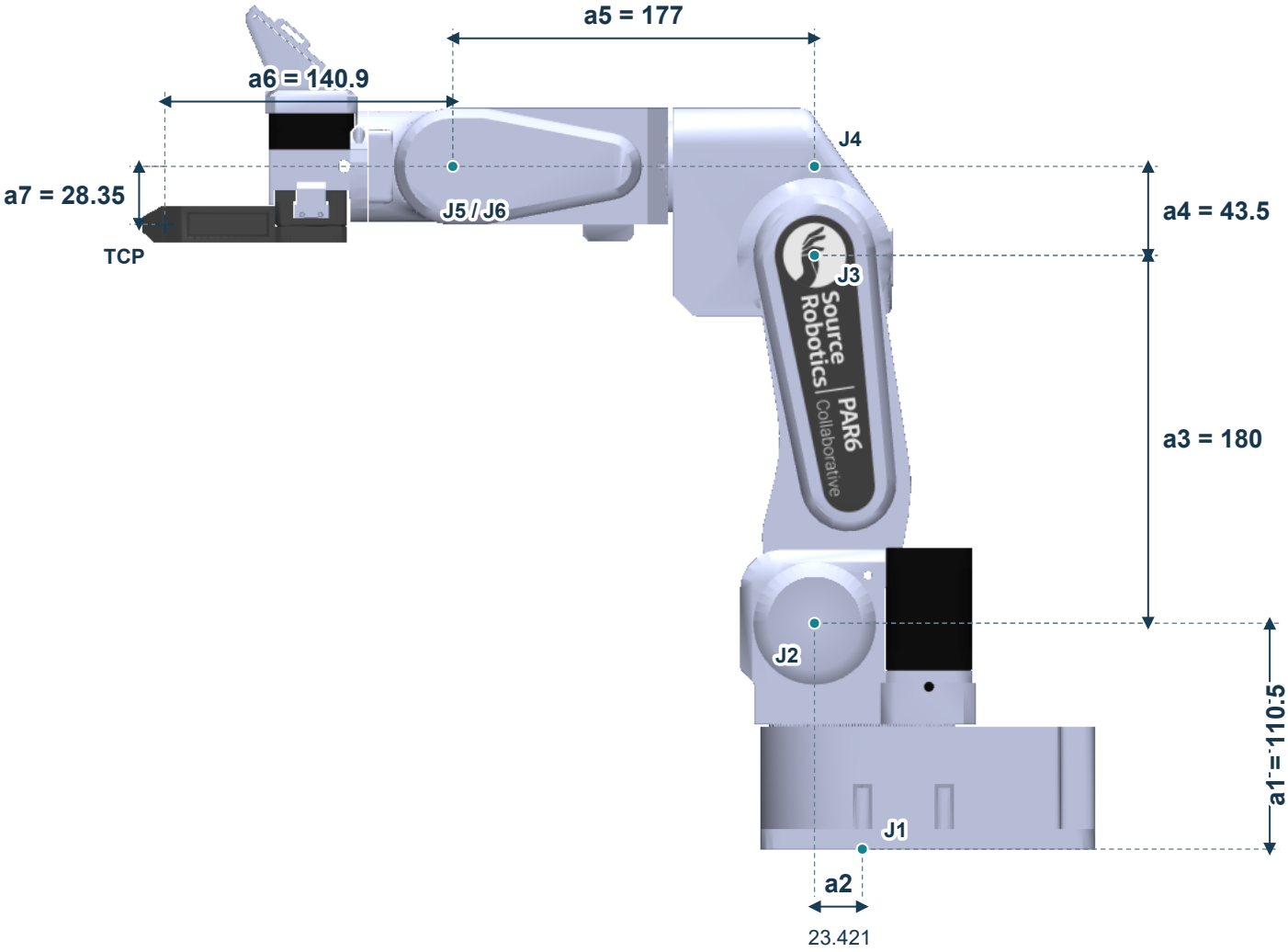
All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

# PAR6 / MSG / Solid jaws

## Kinematic review | Standard DH to the configured TCP



Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.




a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

| Joint | Link | theta [rad] | alpha [deg] | a [mm] | d [mm] |
|---|---|---|---|---|---|
| 1 | Base | q1 | -90 | a2 = 23.421 | a1 = 110.5 |
| 2 | Shoulder | q2 - pi/2 | 180 | a3 = 180 | 0 |
| 3 | Elbow | q3 + pi | 90 | -a4 = -43.5 | 0 |
| 4 | Wrist 1 | q4 | -90 | 0 | -a5 = -177 |
| 5 | Wrist 2 | q5 | 90 | 0 | 0 |
| 6 | Wrist 3 / TCP | q6 + pi | 180 | -a7 = -28.35 | -a6 = -140.9 |

MSG jaw variants share this operational TCP; it is not a measured grasp center.
The 28.35 mm lateral TCP offset is included in DH row 6 as a = -a7.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.001285 mm / 0.000547 deg
Max ideal DH vs URDF: 0.004992 mm / 0.000860 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

Source: par6_msg.urdf + scene_msg.xml | Full transforms, hashes and checks: msg.json

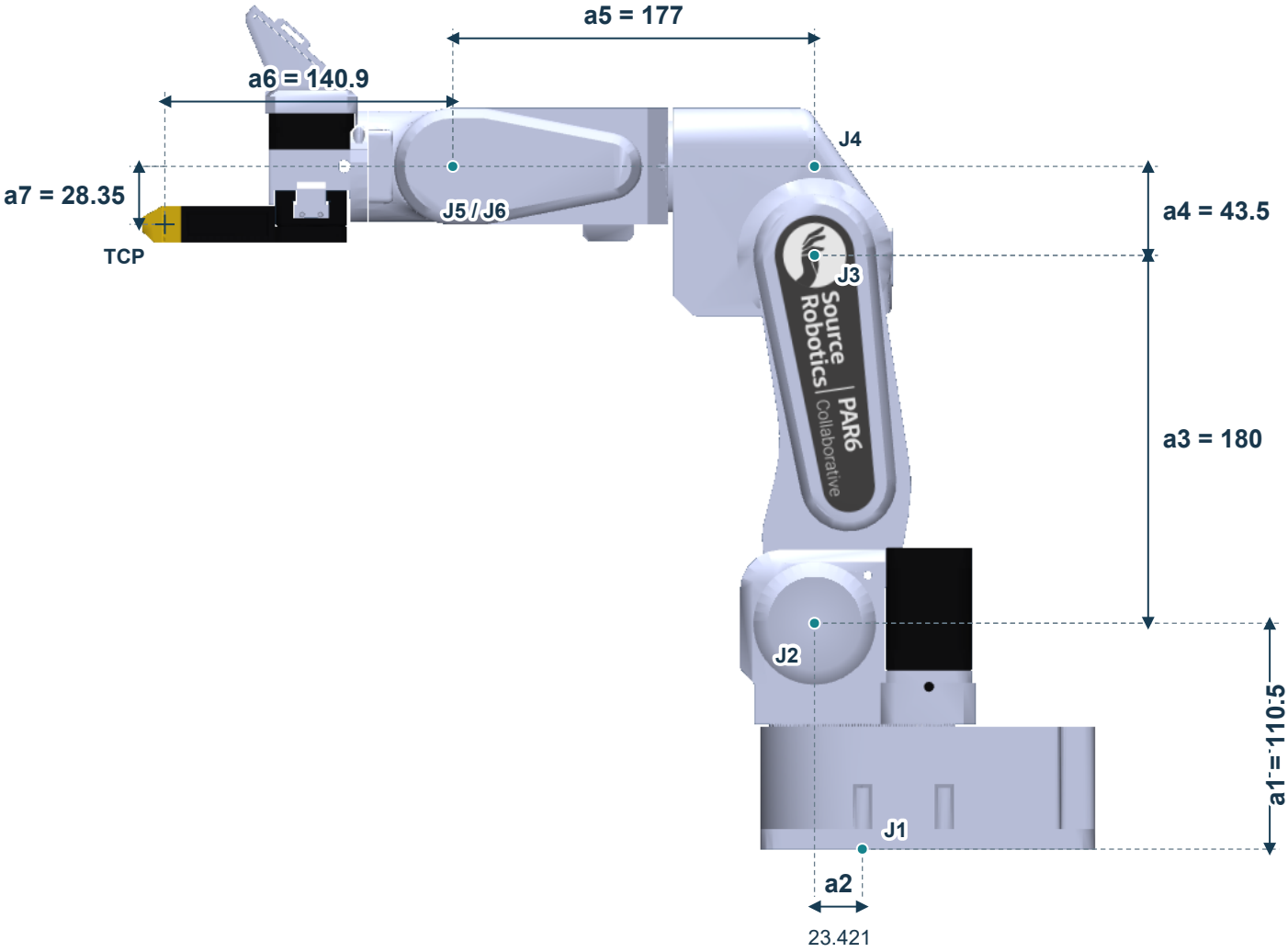
All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

# PAR6 / MSG / Soft TPU jaws

## Kinematic review | Standard DH to the configured TCP



Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.




a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

| Joint | Link | theta [rad] | alpha [deg] | a [mm] | d [mm] |
|---|---|---|---|---|---|
| 1 | Base | q1 | -90 | a2 = 23.421 | a1 = 110.5 |
| 2 | Shoulder | q2 - pi/2 | 180 | a3 = 180 | 0 |
| 3 | Elbow | q3 + pi | 90 | -a4 = -43.5 | 0 |
| 4 | Wrist 1 | q4 | -90 | 0 | -a5 = -177 |
| 5 | Wrist 2 | q5 | 90 | 0 | 0 |
| 6 | Wrist 3 / TCP | q6 + pi | 180 | -a7 = -28.35 | -a6 = -140.9 |

MSG jaw variants share this operational TCP; it is not a measured grasp center.
The 28.35 mm lateral TCP offset is included in DH row 6 as a = -a7.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.001285 mm / 0.000547 deg
Max ideal DH vs URDF: 0.004992 mm / 0.000860 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

Source: par6_msg_tpu_glue.urdf + scene_msg_tpu_glue.xml | Full transforms, hashes and checks: msg_tpu_glue.json

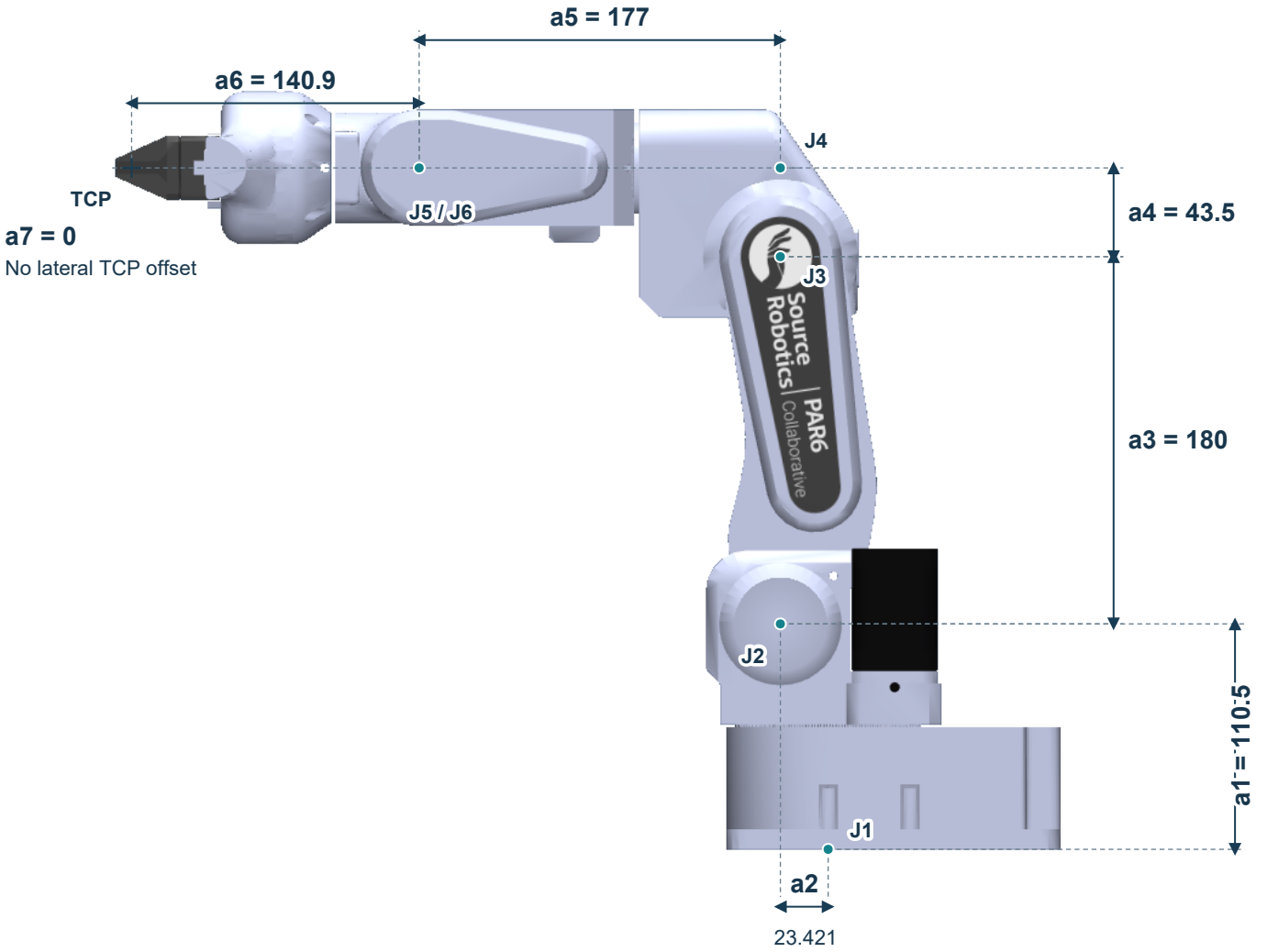
All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

# PAR6 / SSG48 / Solid jaws

## Kinematic review | Standard DH to the configured TCP



Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.




a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

| Joint | Link | theta [rad] | alpha [deg] | a [mm] | d [mm] |
|---|---|---|---|---|---|
| 1 | Base | q1 | -90 | a2 = 23.421 | a1 = 110.5 |
| 2 | Shoulder | q2 - pi/2 | 180 | a3 = 180 | 0 |
| 3 | Elbow | q3 + pi | 90 | -a4 = -43.5 | 0 |
| 4 | Wrist 1 | q4 | -90 | 0 | -a5 = -177 |
| 5 | Wrist 2 | q5 | 90 | 0 | 0 |
| 6 | Wrist 3 / TCP | q6 + pi | 180 | -a7 = 0 | -a6 = -140.9 |

SSG48 jaw variants share this operational TCP; it is not a measured grasp center.
Solid jaws are about 23.4 mm shorter than Fin-ray jaws; their TCP was retained.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.000915 mm / 0.000339 deg
Max ideal DH vs URDF: 0.005262 mm / 0.001048 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

Source: par6_ssg48.urdf + scene_ssg48.xml | Full transforms, hashes and checks: ssg48.json

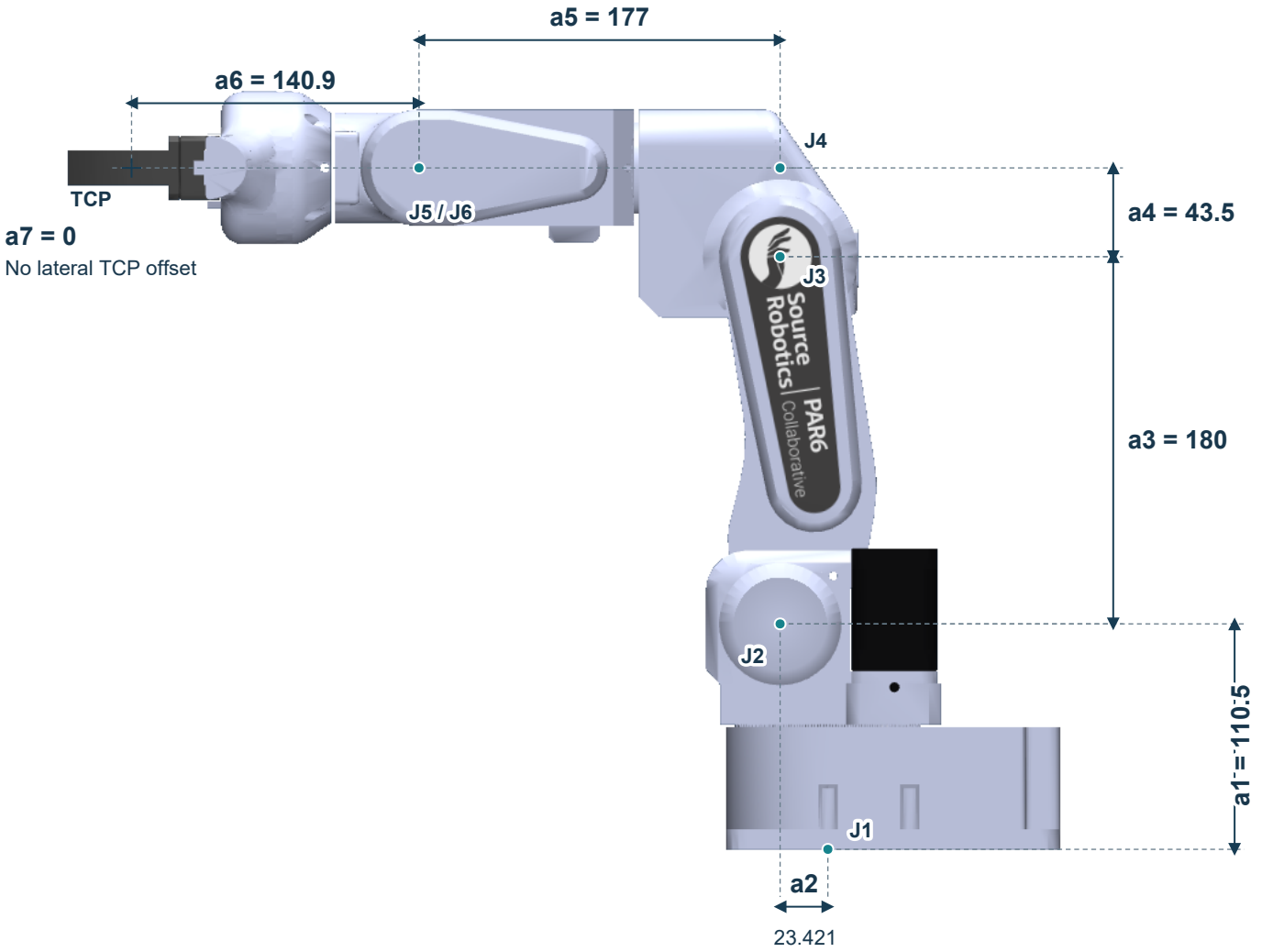
All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

# PAR6 / SSG48 / Fin-ray jaws

## Kinematic review | Standard DH to the configured TCP



Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.




a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

| Joint | Link | theta [rad] | alpha [deg] | a [mm] | d [mm] |
|---|---|---|---|---|---|
| 1 | Base | q1 | -90 | a2 = 23.421 | a1 = 110.5 |
| 2 | Shoulder | q2 - pi/2 | 180 | a3 = 180 | 0 |
| 3 | Elbow | q3 + pi | 90 | -a4 = -43.5 | 0 |
| 4 | Wrist 1 | q4 | -90 | 0 | -a5 = -177 |
| 5 | Wrist 2 | q5 | 90 | 0 | 0 |
| 6 | Wrist 3 / TCP | q6 + pi | 180 | -a7 = 0 | -a6 = -140.9 |

SSG48 jaw variants share this operational TCP; it is not a measured grasp center.
Solid jaws are about 23.4 mm shorter than Fin-ray jaws; their TCP was retained.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.000915 mm / 0.000339 deg
Max ideal DH vs URDF: 0.005262 mm / 0.001048 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

Source: par6_ssg48_stock.urdf + scene_ssg48_stock.xml | Full transforms, hashes and checks: ssg48_stock.json

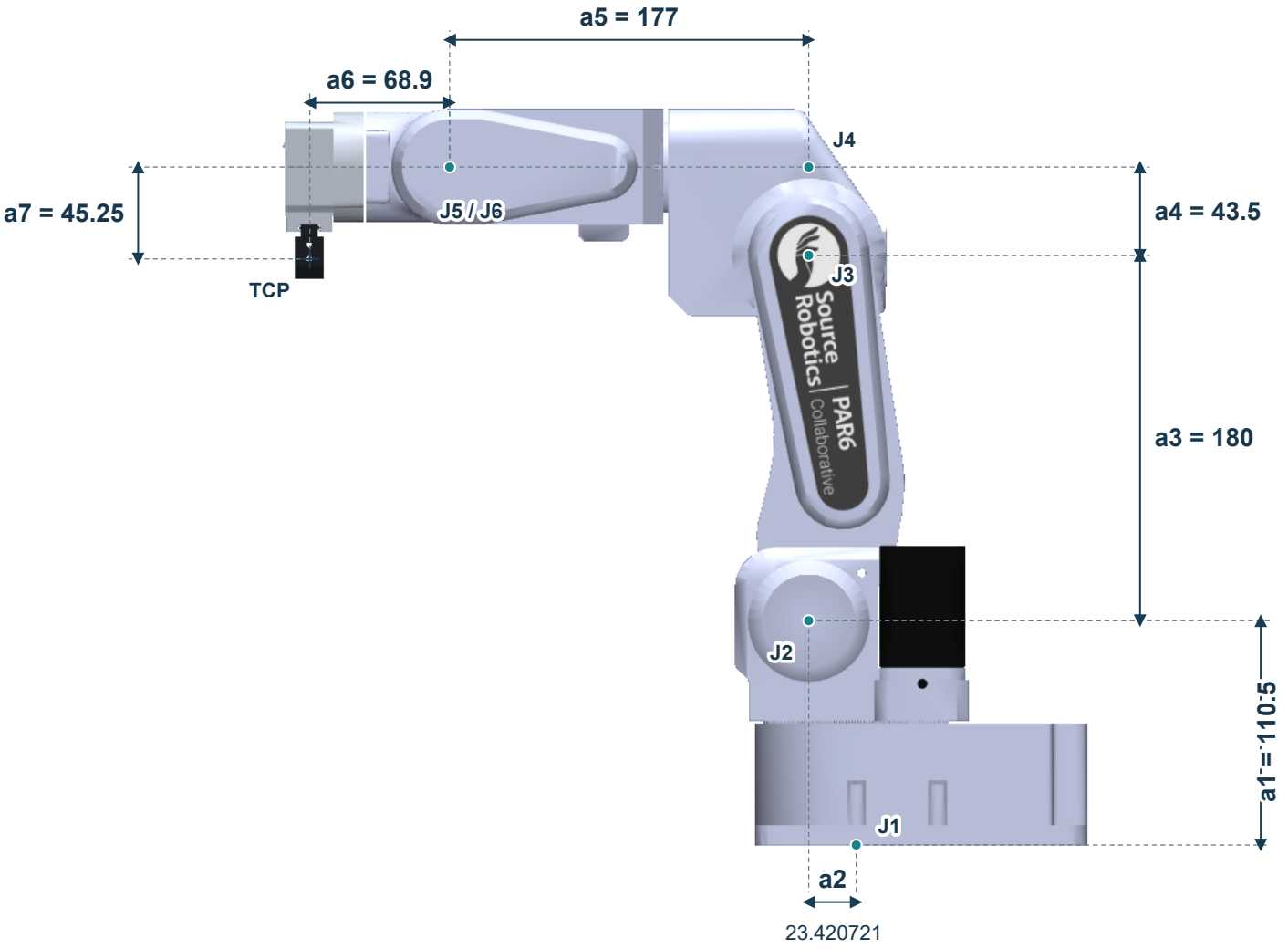
All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

# PAR6 / Pneumatic gripper

## Kinematic review | Standard DH to the configured TCP



Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.




a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

| Joint | Link | theta [rad] | alpha [deg] | a [mm] | d [mm] |
|---|---|---|---|---|---|
| 1 | Base | q1 | -90 | a2 = 23.420721 | a1 = 110.5 |
| 2 | Shoulder | q2 - pi/2 | 180 | a3 = 180 | 0 |
| 3 | Elbow | q3 + pi | 90 | -a4 = -43.5 | 0 |
| 4 | Wrist 1 | q4 | -90 | 0 | -a5 = -177 |
| 5 | Wrist 2 | q5 | 90 | 0 | 0 |
| 6 | Wrist 3 / TCP | q6 + pi | 180 | -a7 = -45.25 | -a6 = -68.9 |

Pneumatic TCP follows the legacy MHZ2-10D reference; it is not a newly measured grasp center.
Jaws close symmetrically through 3.5 mm each. Mass and finite actuator settings are estimates.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.004340 mm / 0.001350 deg
Max ideal DH vs URDF: 0.000000 mm / 0.000000 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

Source: par6_pneumatic.urdf + scene_pneumatic.xml | Full transforms, hashes and checks: pneumatic.json

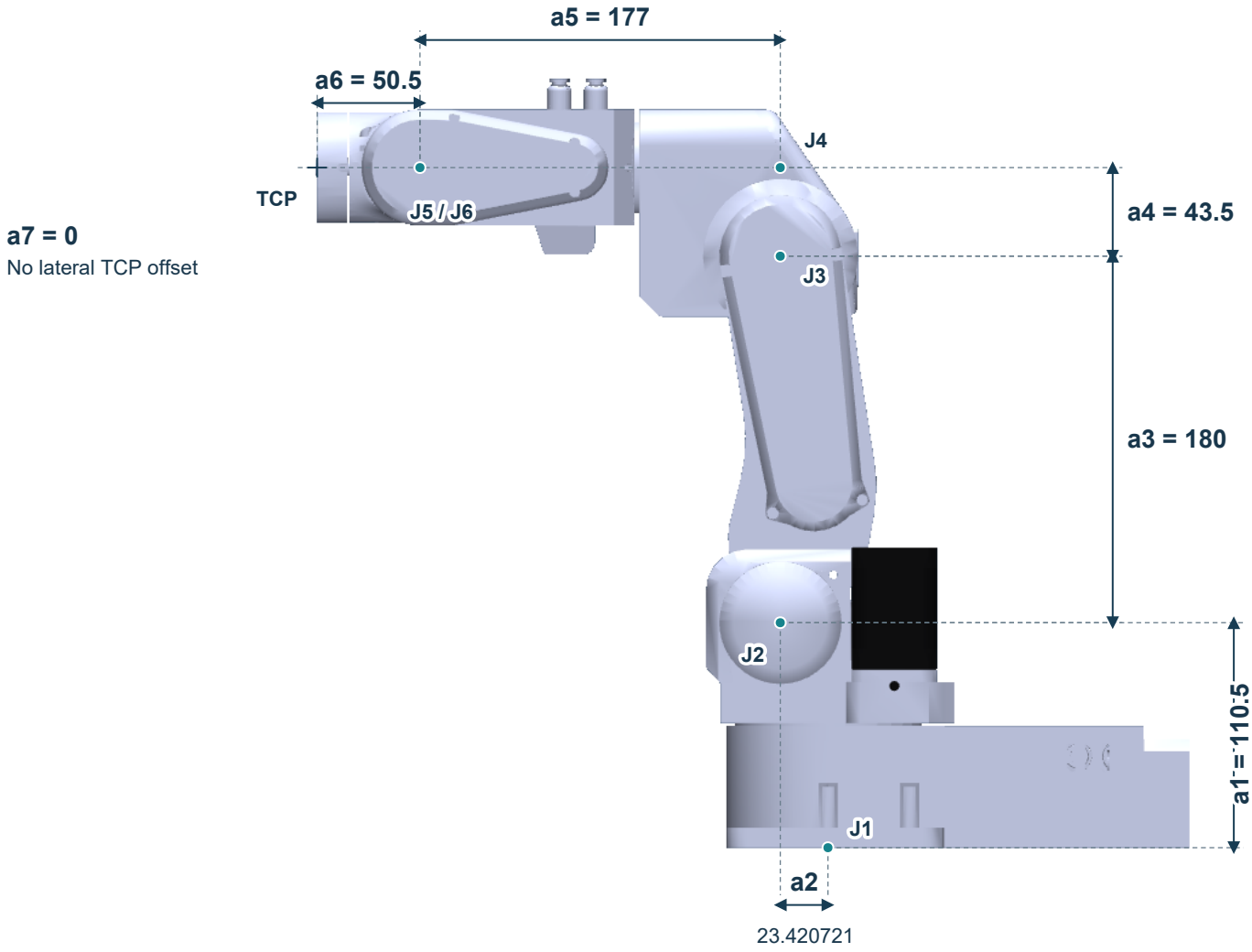
All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

# PAROL6 / Flange only

## Kinematic review | Standard DH to the configured TCP



Pose: q1 ... q6 = 0 rad; no jaws. Orthographic side view; dimensions in mm.




a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

| Joint | Link | theta [rad] | alpha [deg] | a [mm] | d [mm] |
|---|---|---|---|---|---|
| 1 | Base | q1 | -90 | a2 = 23.420721 | a1 = 110.5 |
| 2 | Shoulder | q2 - pi/2 | 180 | a3 = 180 | 0 |
| 3 | Elbow | q3 + pi | 90 | -a4 = -43.5 | 0 |
| 4 | Wrist 1 | q4 | -90 | 0 | -a5 = -177 |
| 5 | Wrist 2 | q5 | 90 | 0 | 0 |
| 6 | Wrist 3 / TCP | q6 + pi | 180 | -a7 = 0 | -a6 = -50.5 |

TCP is the configured flange reference. No gripper is installed.
a6 measures J6 center to flange; a7 is zero.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.002206 mm / 0.001026 deg
Max ideal DH vs URDF: 0.000000 mm / 0.000000 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

Source: parol6_flange.urdf + scene_flange.xml | Full transforms, hashes and checks: flange.json

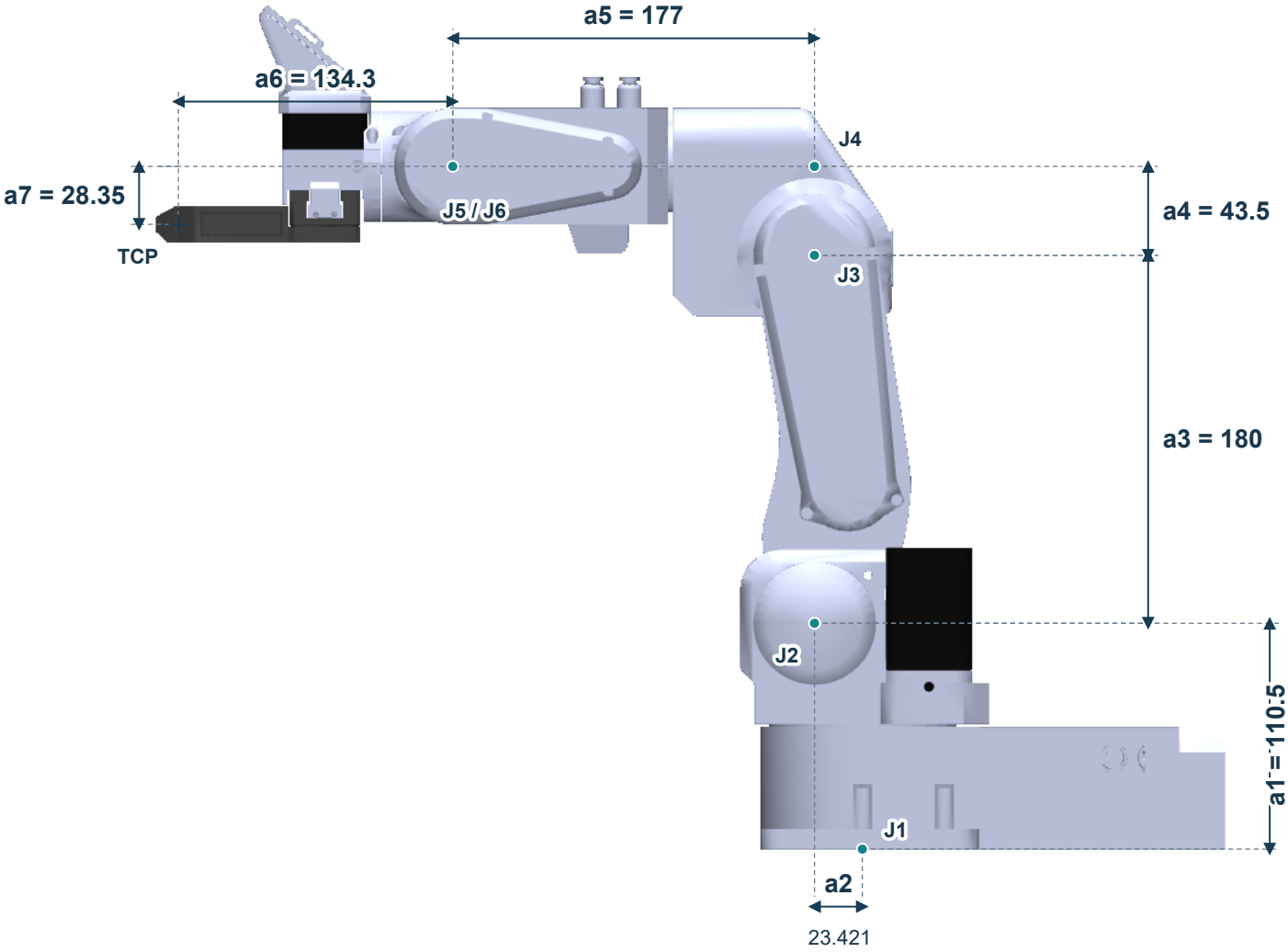
All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

# PAROL6 / MSG / Solid jaws

## Kinematic review | Standard DH to the configured TCP



Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.




a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

| Joint | Link | theta [rad] | alpha [deg] | a [mm] | d [mm] |
|---|---|---|---|---|---|
| 1 | Base | q1 | -90 | a2 = 23.421 | a1 = 110.5 |
| 2 | Shoulder | q2 - pi/2 | 180 | a3 = 180 | 0 |
| 3 | Elbow | q3 + pi | 90 | -a4 = -43.5 | 0 |
| 4 | Wrist 1 | q4 | -90 | 0 | -a5 = -177 |
| 5 | Wrist 2 | q5 | 90 | 0 | 0 |
| 6 | Wrist 3 / TCP | q6 + pi | 180 | -a7 = -28.35 | -a6 = -134.3 |

MSG jaw variants share this operational TCP; it is not a measured grasp center.
The 28.35 mm lateral TCP offset is included in DH row 6 as a = -a7.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.001000 mm / 0.000479 deg
Max ideal DH vs URDF: 0.002787 mm / 0.000679 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

Source: parol6_msg.urdf + scene_msg.xml | Full transforms, hashes and checks: msg.json

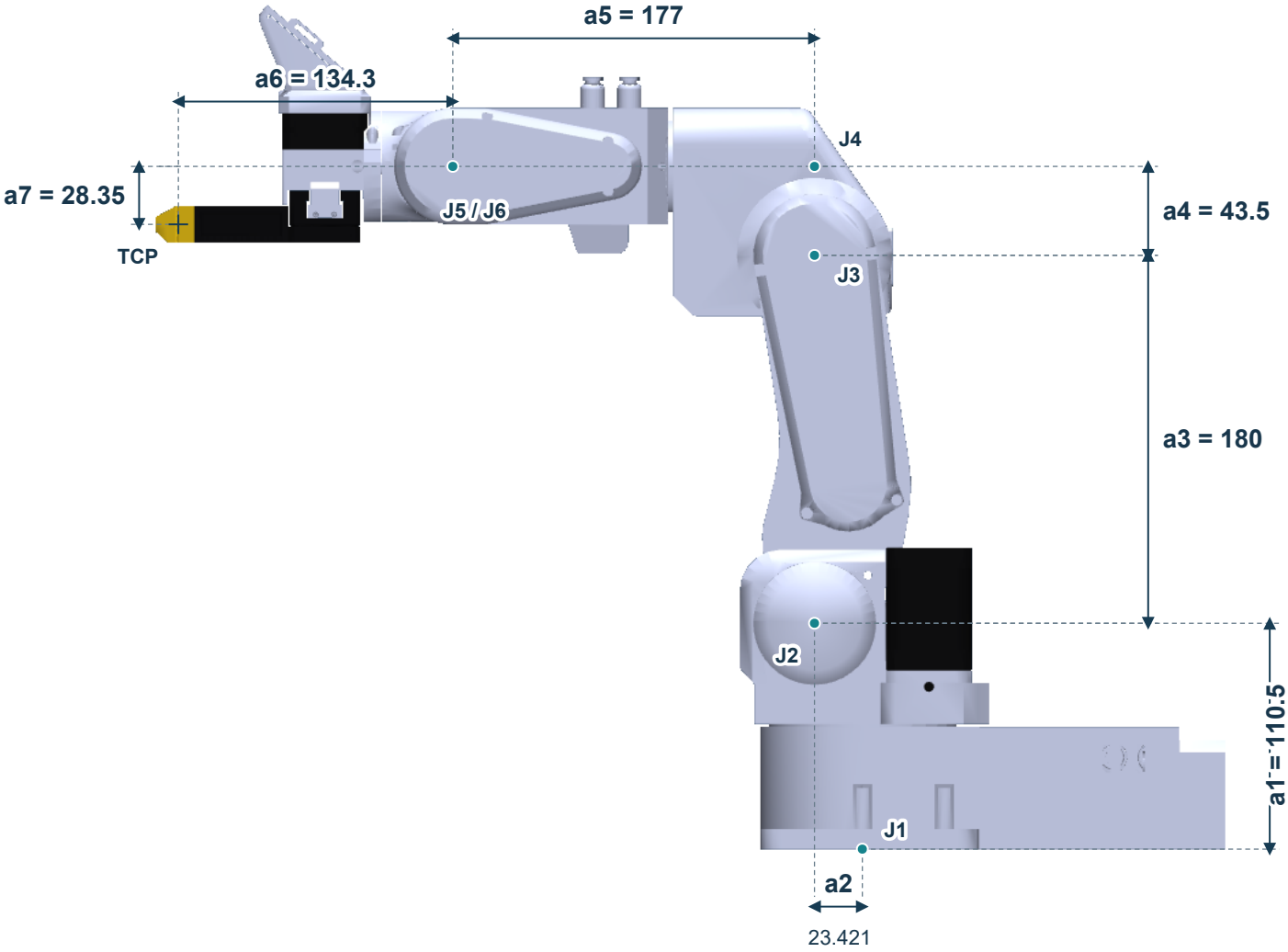
All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

# PAROL6 / MSG / Soft TPU jaws

## Kinematic review | Standard DH to the configured TCP



Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.




a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

| Joint | Link | theta [rad] | alpha [deg] | a [mm] | d [mm] |
|---|---|---|---|---|---|
| 1 | Base | q1 | -90 | a2 = 23.421 | a1 = 110.5 |
| 2 | Shoulder | q2 - pi/2 | 180 | a3 = 180 | 0 |
| 3 | Elbow | q3 + pi | 90 | -a4 = -43.5 | 0 |
| 4 | Wrist 1 | q4 | -90 | 0 | -a5 = -177 |
| 5 | Wrist 2 | q5 | 90 | 0 | 0 |
| 6 | Wrist 3 / TCP | q6 + pi | 180 | -a7 = -28.35 | -a6 = -134.3 |

MSG jaw variants share this operational TCP; it is not a measured grasp center.
The 28.35 mm lateral TCP offset is included in DH row 6 as a = -a7.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.001000 mm / 0.000479 deg
Max ideal DH vs URDF: 0.002787 mm / 0.000679 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

Source: parol6_msg_tpu_glue.urdf + scene_msg_tpu_glue.xml | Full transforms, hashes and checks: msg_tpu_glue.json

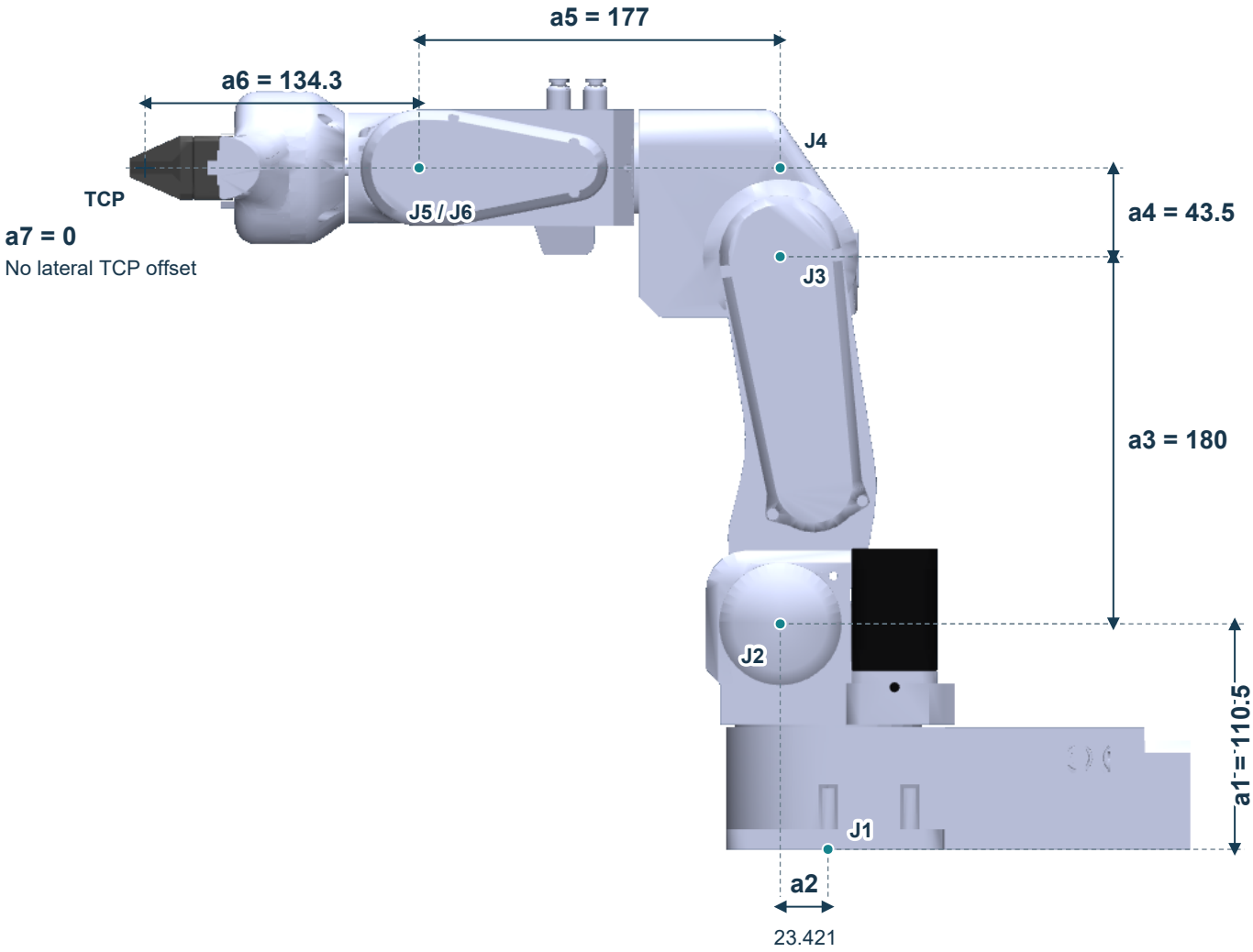
All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

# PAROL6 / SSG48 / Solid jaws

## Kinematic review | Standard DH to the configured TCP



Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.




a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

| Joint | Link | theta [rad] | alpha [deg] | a [mm] | d [mm] |
|---|---|---|---|---|---|
| 1 | Base | q1 | -90 | a2 = 23.421 | a1 = 110.5 |
| 2 | Shoulder | q2 - pi/2 | 180 | a3 = 180 | 0 |
| 3 | Elbow | q3 + pi | 90 | -a4 = -43.5 | 0 |
| 4 | Wrist 1 | q4 | -90 | 0 | -a5 = -177 |
| 5 | Wrist 2 | q5 | 90 | 0 | 0 |
| 6 | Wrist 3 / TCP | q6 + pi | 180 | -a7 = 0 | -a6 = -134.3 |

SSG48 jaw variants share this operational TCP; it is not a measured grasp center.
Solid jaws are about 23.4 mm shorter than Fin-ray jaws; their TCP was retained.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.000818 mm / 0.000316 deg
Max ideal DH vs URDF: 0.002957 mm / 0.000764 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

Source: parol6_ssg48.urdf + scene_ssg48.xml | Full transforms, hashes and checks: ssg48.json

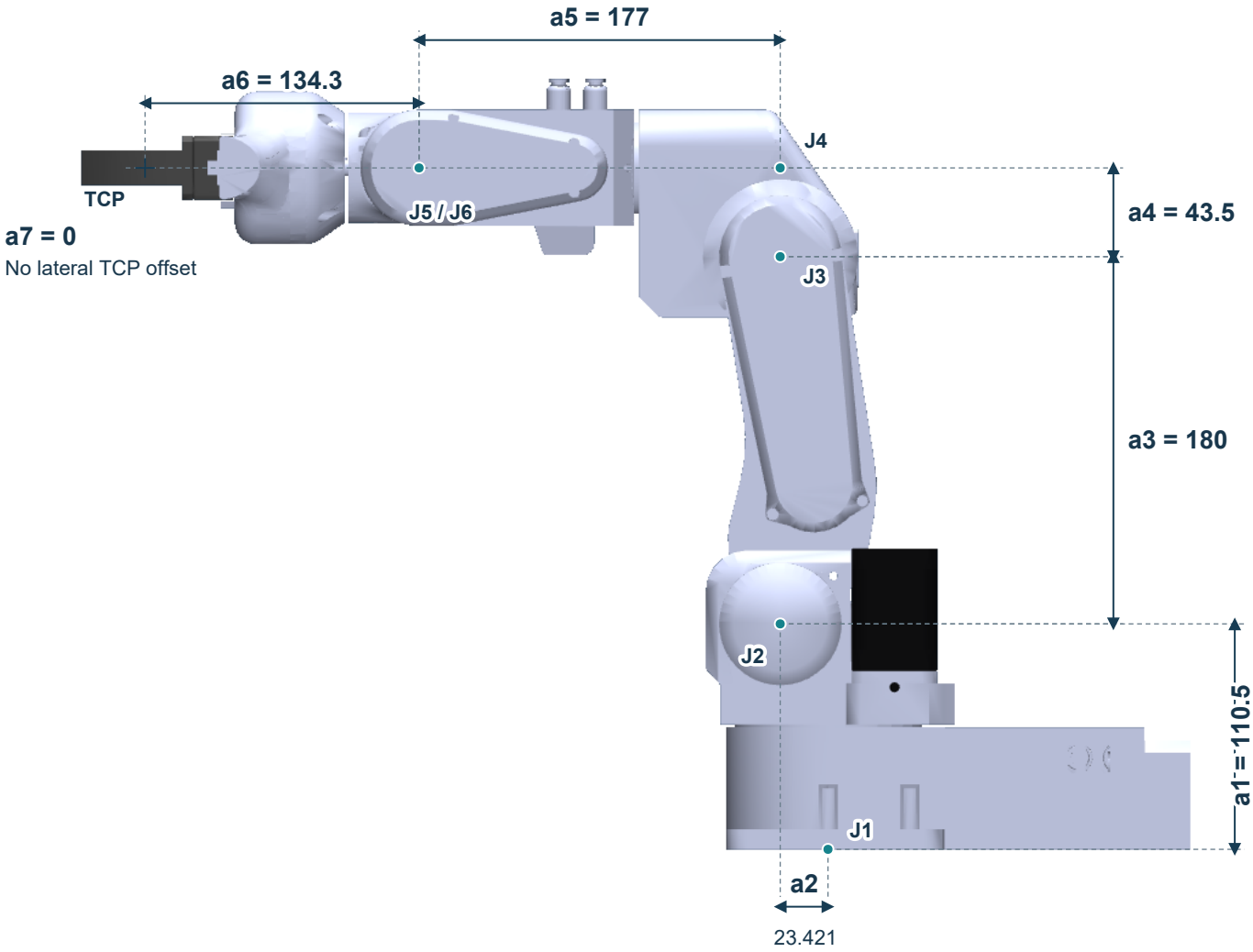
All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

# PAROL6 / SSG48 / Fin-ray jaws

## Kinematic review | Standard DH to the configured TCP



Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.




a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

| Joint | Link | theta [rad] | alpha [deg] | a [mm] | d [mm] |
|---|---|---|---|---|---|
| 1 | Base | q1 | -90 | a2 = 23.421 | a1 = 110.5 |
| 2 | Shoulder | q2 - pi/2 | 180 | a3 = 180 | 0 |
| 3 | Elbow | q3 + pi | 90 | -a4 = -43.5 | 0 |
| 4 | Wrist 1 | q4 | -90 | 0 | -a5 = -177 |
| 5 | Wrist 2 | q5 | 90 | 0 | 0 |
| 6 | Wrist 3 / TCP | q6 + pi | 180 | -a7 = 0 | -a6 = -134.3 |

SSG48 jaw variants share this operational TCP; it is not a measured grasp center.
Solid jaws are about 23.4 mm shorter than Fin-ray jaws; their TCP was retained.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.000818 mm / 0.000316 deg
Max ideal DH vs URDF: 0.002957 mm / 0.000764 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

Source: parol6_ssg48_stock.urdf + scene_ssg48_stock.xml | Full transforms, hashes and checks: ssg48_stock.json

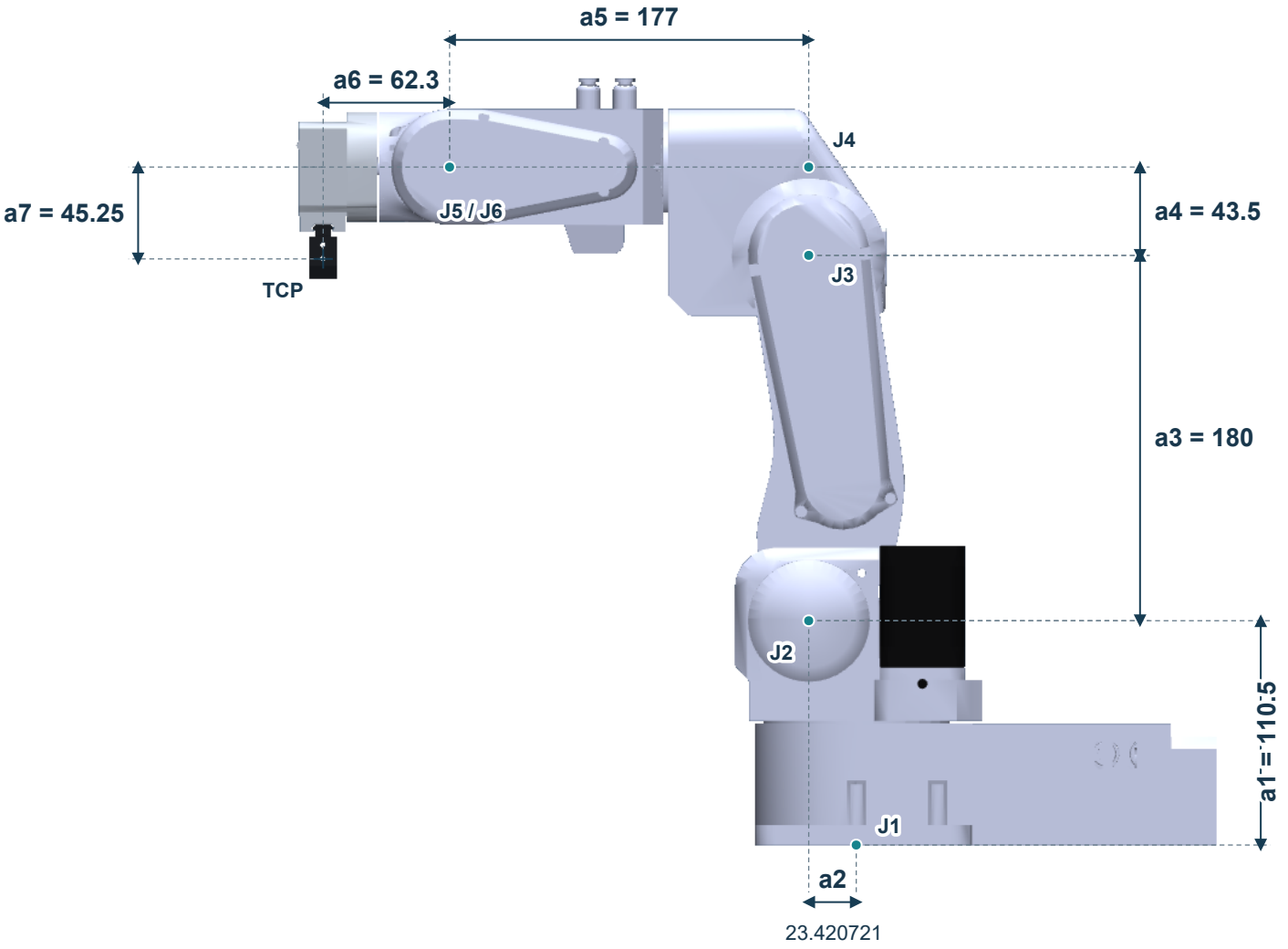
All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.

# PAROL6 / Pneumatic gripper

## Kinematic review | Standard DH to the configured TCP



Pose: q1 ... q6 = 0 rad; jaws open. Orthographic side view; dimensions in mm.




a1 Base height a2 Shoulder offset a3 Upper arm a4 Elbow offset
a5 Forearm axis distance a6 J6-to-TCP axial reach a7 TCP lateral offset

A_i = Rz(theta_i) Tz(d_i) Tx(a_i) Rx(alpha_i)
q = model joint angles (rad). a = link length along X; d = link offset along Z.

| Joint | Link | theta [rad] | alpha [deg] | a [mm] | d [mm] |
|---|---|---|---|---|---|
| 1 | Base | q1 | -90 | a2 = 23.420721 | a1 = 110.5 |
| 2 | Shoulder | q2 - pi/2 | 180 | a3 = 180 | 0 |
| 3 | Elbow | q3 + pi | 90 | -a4 = -43.5 | 0 |
| 4 | Wrist 1 | q4 | -90 | 0 | -a5 = -177 |
| 5 | Wrist 2 | q5 | 90 | 0 | 0 |
| 6 | Wrist 3 / TCP | q6 + pi | 180 | -a7 = -45.25 | -a6 = -62.3 |

Pneumatic TCP follows the legacy MHZ2-10D reference; it is not a newly measured grasp center.
Jaws close symmetrically through 3.5 mm each. Mass and finite actuator settings are estimates.

Independent FK checks: 128 poses, including zero and joint-limit bounds.
Max URDF vs MuJoCo TCP: 0.002447 mm / 0.001026 deg
Max ideal DH vs URDF: 0.000000 mm / 0.000000 deg
Right-angle DH approximates rounded export rotations. The snapshot uses the MuJoCo visual meshes.

Source: parol6_pneumatic.urdf + scene_pneumatic.xml | Full transforms, hashes and checks: pneumatic.json

All sheets use the same scale. Side projection: +X points left, +Z up; view from +Y.